The Tactical Substrate

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P1 — THE TACTICAL SUBSTRATE

Series: The Implications of Edge Degraded Ops **Paper:** P1 of 11 **Version:** v0.1-seed **Date:** 2026-04-30 **Author:** Yeti (nonsequitur.tech) **DOI:** pending Zenodo deposit

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ABSTRACT

The C5ISR edge is not an extension of the enterprise network. It is a fundamentally different operating environment where the WAN link is intermittent by design, blackout windows are not anomalies but the expected envelope, and session-oriented protocols fail structurally rather than occasionally. This paper — P1 of *The Implications of Edge Degraded Ops* — provides the survey frame. It names the substrate: Delay-Tolerant Networking bundle custody, write-ahead-log replication, CRDT-based state coherence, and an out-of-path AI Supervisor operating under the HGC^3AE^2 human-governance constraint. Nine problem-unpack papers (P3–P11) follow this frame; P2 provides the governor treatment. A senior architect who reads only this paper can defend the architecture at a technical briefing. Every other paper in the series is the detailed argument for one of the claims made here.

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§1 — THE OPERATIONAL PICTURE: WHY THE EDGE IS NOT THE ENTERPRISE

The scenario is not hypothetical. A naval surface combatant operating in a contested littoral environment maintains a satellite uplink that is periodically eclipsed by terrain, jammed by adversary electronic warfare, or simply unavailable during emissions-control periods when the ship goes dark by doctrine. A forward operating base in a mountainous theater relies on a BLOS radio link that cycles through connectivity windows measured in minutes, not hours. A tactical operations center embedded with a ground force unit communicates through a PACE plan — Primary, Alternate, Contingency, Emergency — where the primaries fail routinely and the contingency channel is a low-rate HF connection that can carry text but not a database transaction. In each of these scenarios, the WAN link is not a constant that the architecture can assume. It is a variable — intermittent, bandwidth-constrained, latency-asymmetric, and subject to interruption without warning.

C5ISR systems — Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance — must exchange situational awareness data, targeting solutions, sensor feeds, and mission-coordination packets across this link. The tempo of modern combined-arms operations does not pause for link availability. A targeting cycle that requires end-to-end session persistence for its data synchronization will simply fail when the link drops mid-exchange. The question is not whether it fails but what the failure mode looks like in practice and whether the architecture was designed to contain it or merely tolerate it.

What breaks, and in what order: when a TCP session is interrupted mid-transfer — whether by link dropout, RF interference, or emissions control — the session state evaporates. The TCP state machine at each end times out; the application layer receives a connection-reset or read-error event; the operator sees a hung interface or an error dialog. If the application has no persistent queue below the session layer, the partially transferred payload is lost or must be retransmitted from scratch when connectivity restores. In a fire control system, that means a targeting solution that was 70% complete must be rebuilt. In a sensor-feed aggregator, it means a gap in the intelligence picture that may not be flaggable — the operator cannot distinguish between “no data” and “no link.” In a logistics system, it means a resupply request that appeared to send but never arrived at the receiving node. These are not edge cases; they are the routine operational experience of units operating at the tactical edge.

WAN optimization tools — forward error correction, protocol acceleration, data deduplication at the link layer — address the problem at the wrong level. They optimize the behavior of the link when it exists. They have no model for the link’s absence. A WAN optimizer cannot custody a bundle of data and deliver it when the link next opens. It cannot merge divergent state that accumulated on both sides during a blackout window. It cannot make a routing decision based on predicted link availability from historical contact graphs. WAN optimization assumes the session; the tactical edge invalidates that assumption structurally. The problem is not link quality — it is link continuity, and no amount of forward error correction resolves a link that is not there.

Quorum-based consensus protocols — Raft, Paxos, and their variants — fail in the same structural way for a different reason. Quorum requires a majority of participants to be reachable for any write to commit. In a two-node ship-to-shore topology with a blackout window, there is no quorum. The shore-side node cannot advance its log; the ship-side node cannot commit a write that the shore side will confirm. If the system is designed for strict consistency, it deadlocks. If it degrades gracefully, it requires explicit partition-tolerance engineering — which is exactly what standard Raft implementations do not provide for asymmetric DDIL topologies. The problem is not that consensus is wrong. The problem is that consensus assumes a network that the tactical edge cannot guarantee.

The thesis of this series follows directly from this failure analysis. The tactical substrate must be designed for the link’s absence, not its presence. DTN bundle custody, write-ahead-log replication, and an out-of-path AI Supervisor give the architecture that posture. Bundle custody guarantees that a data unit accepted by a DTN agent will be delivered exactly once when a contact window opens, regardless of how long that window takes to arrive. Write-ahead logging provides a durable, ordered record of state transitions that can be replayed or merged after a blackout without quorum. The AI Supervisor, operating out of the data path and never in the write path, monitors link state and adjusts routing policy, custody priorities, and merge strategies within the envelope of human-set thresholds — it does not govern, it executes. Together these three mechanisms transform a link-dependent architecture into a substrate that treats connectivity as a resource to be scheduled rather than an assumption to be made.

Figure 1 — Operational Picture: Ship-Shore-FOB DDIL Topology *Illustrates the canonical three-node tactical topology — naval combatant, forward operating base, shore command — with annotated link characteristics: satellite window duration, RF blackout periods, BLOS channel rate, and emissions-control constraints. Shows the C5ISR data flows that must cross these links and the failure modes each link type produces.* [FIGURE SEED: assets/series-3-edo/figures-seed/slide-02.png — full SVG redraw at v1.0-preprint]

§2 — WHY TCP-ANYTHING FAILS; WHY WAN OPTIMIZATION IS THE WRONG LAYER

Thesis: TCP's session model and WAN optimization's link-tuning approach both assume the existence of an end-to-end path. At the tactical edge, the path's absence is not an exception to handle — it is the operating condition the architecture must be built around. The failure is structural and cannot be remedied by tuning.

TCP session death on link loss. A TCP connection is a state machine with a finite timeout. When the underlying link disappears, the session state on both endpoints begins expiring. No application-layer retry logic recovers the mid-flight payload; it must be retransmitted from the beginning when connectivity restores. In high-latency SATCOM environments, retransmission penalty compounds severely.

WAN optimization assumes the session. Protocol accelerators, header compression, data deduplication, and forward error correction all operate on traffic flowing through an active link. They have no mechanism for store-and-forward across a blackout window. A WAN optimizer cannot custody a partially transferred payload and resume delivery when the link opens.

No quorum in blackout. Raft, Paxos, and multi-Paxos all require a reachable majority to commit a write. In a two-node or asymmetric partition — ship isolated from shore — the write cannot commit under strict consistency. The system either halts or requires partition-tolerance engineering that standard consensus libraries do not provide at DDIL topology.

Latency asymmetry breaks TCP congestion control. SATCOM links exhibit asymmetric latency profiles and high bandwidth-delay products that defeat TCP's AIMD congestion control. Even when the link is up, TCP throughput can be a fraction of available bandwidth without specialized tuning — and the tuning addresses only the link-present case.

The wrong layer for the problem. WAN optimization is a link-layer concern. The tactical substrate problem is a data-lifecycle concern: who holds the data, in what form, for how long, and how does state converge across nodes that were partitioned. That is an application-layer architectural question that no link-layer tool can answer.

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§3 — THE SUBSTRATE STACK: DTN → WAL → CRDT → AI SUPERVISOR

Thesis: Four mechanisms, layered in order of custody: DTN bundle protocol handles store-and-forward at the network layer; the write-ahead log provides durable ordered state at the data layer; CRDTs provide convergent state merging above the log; and the AI Supervisor operates out of path, adjusting policy without touching the data plane. The stack is not a framework — it is a set of composable primitives, each solving the narrowest problem it can.

DTN (Delay-Tolerant Networking / RFC 5050 [@RFC5050]). Bundle custody ensures that a data unit accepted by a DTN agent will be delivered exactly once when a contact window opens. Custody transfer moves responsibility from the originating node to the next hop as each hop is traversed. No end-to-end path required at the time of send. Contact plans — scheduled or opportunistic — drive routing decisions.

WAL (Write-Ahead Log). Every state transition is logged durably before it is applied. The log is the source of truth for replay after a blackout, for audit, and for the CRDT merge input. WAL is the catalog layer in HGC^3AE^2 C^3 terms — it is what curation operates on.

CRDT (Conflict-free Replicated Data Type). For state that must converge across nodes that operated independently during a blackout, CRDTs provide a mathematically guaranteed merge without a coordinator. The AI Supervisor does not adjudicate CRDT merges — the type semantics do. The Supervisor monitors merge outcomes and escalates to human decision when merge results exceed drift thresholds.

AI Supervisor (out-of-path). Monitors link state, WAL lag, CRDT drift, and custody queue depth. Adjusts routing policy and custody priorities within human-set bounds. Does not write to the data path; does not issue custody transfers; does not resolve merge conflicts. Executes inside the HGC^3AE^2 H-governs / AE^2-enforces envelope. Detailed in §5 and P5 (Orchestration Problem).

Figure 2 — Substrate Stack Overview *Four-layer diagram: DTN bundle layer, WAL data layer, CRDT state layer, AI Supervisor control plane (shown out-of-path). Annotations indicate the data flow direction, the custody handoff points, and the human decision gates at the top of the H half.* [FIGURE SEED: assets/series-3-edo/figures-seed/slide-10.png — full SVG redraw at v1.0-preprint]

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§4 — STRETCHING THE NAS: THE LOAD-BEARING PATTERN

Thesis: The NAS (Network-Attached Storage) at the tactical edge is not a passive file server — it is the custody anchor, the WAL host, and the synchronization surface. “Stretching the NAS” is the architectural pattern that makes commodity storage infrastructure into a DDIL-tolerant state node without replacing it. The pattern is load-bearing throughout this series and is unpacked in P4 (State Coherence Problem).

The NAS as WAL host. Write-ahead log segments live on NAS-backed storage. The log survives node restarts, link drops, and power cycles. The NAS is not a synchronization mechanism — the WAL is. The NAS is the durable substrate the WAL writes to.

Custody anchor. When a DTN bundle is accepted by a node, the bundle payload is staged to NAS-backed storage before custody acknowledgment is sent. This ensures that a custody acknowledgment is not issued for data that could be lost on a node failure before the next contact window.

CRDT merge surface. After a blackout, both sides of the partition have accumulated state. The NAS at each node holds the local WAL segments that represent that accumulated state. The CRDT merge operation reads from both WAL surfaces and produces the converged state. No coordinator required; no quorum required.

Commodity hardware, accreditation-ready. The pattern intentionally uses NAS hardware that is available in DoD supply chains, can be ruggedized for shipboard or ground vehicle deployment, and can be accredited under existing RMF control families. The architecture does not require novel hardware.

Unpacked in P4. The full treatment of WAL-on-NAS topology, log segment management, CRDT merge scheduling, and failure mode analysis is in P4 — State Coherence Problem. This paper names the pattern; P4 is where the engineer builds it.

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§5 — THE AI SUPERVISOR: OUT-OF-PATH, CONTROL PLANE ONLY

Thesis: The AI Supervisor is the observability and policy-adjustment layer of the substrate. It operates exclusively in the control plane — it reads telemetry, adjusts parameters, and escalates to human decision when bounds are exceeded. The red line is absolute: the AI Supervisor does not write to the data path, does not issue custody acknowledgments, and does not resolve state conflicts. It executes within human-set policy; it does not set policy.

Observability without authority. The Supervisor subscribes to WAL lag metrics, DTN custody queue depth, CRDT drift vectors, and link-state telemetry. It aggregates these into a substrate health model. It does not act on that model except to adjust parameters that human operators have pre-authorized it to adjust.

Parameter adjustment within bounds. Pre-authorized adjustments include: routing priority weights for custody queues, contact-plan scoring inputs, and CRDT merge-trigger thresholds. Every adjustment is logged. Adjustments that would breach a human-set bound trigger an escalation event — a notification to a human operator with a proposed action and a decision window.

The red line: no data-path writes. The Supervisor does not custody data, does not acknowledge bundles, does not write WAL segments, and does not adjudicate CRDT merges. If the Supervisor's data-path channel were severed, the substrate continues to function — custody, logging, and merge proceed on their own. The Supervisor is supervisory, not load-bearing in the data plane.

HGC³AE² posture. The Supervisor is an AE² enforcement surface — it monitors and enforces, within the envelope set by the H half (human operators). It does not govern. Policy changes require human initiation. The Supervisor executes those policies at machine speed.

Unpacked in P5. The Orchestration Problem paper (P5) provides the full Supervisor design: event taxonomy, escalation protocol, boundary conditions, and the implementation reference for the control-plane interface.

Figure 3 — AI Supervisor Out-of-Path Architecture *Diagram showing the substrate data path (DTN → WAL → CRDT merge → application layer) with the AI Supervisor positioned off to the side, connected by read-only telemetry feeds and write-only parameter-adjustment channels. The red line between Supervisor and data path is annotated explicitly. Human decision gate shown at the top of the escalation chain.* [FIGURE SEED: assets/series-3-edo/figures-seed/slide-09.png — full SVG redraw at v1.0-preprint]

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§6 — THREE OPERATIONAL MODES: CONNECTED / DEGRADED / BLACKOUT

Thesis: The substrate is not a single-mode system — it runs in three explicitly designed operational modes, each with its own synchronization posture, custody behavior, and AI Supervisor operating bounds. Mode transitions are human-authorized by default; the Supervisor can recommend a transition, not execute one.

Connected mode. Link available, latency within bounds, full synchronization. WAL segments replicate in near-real-time. CRDT merges are routine and low-drama. DTN bundles are delivered in seconds to minutes. AI Supervisor operates in steady-state monitoring posture with wide parameter bounds.

Degraded mode. Link intermittent or bandwidth-constrained. WAL replication falls behind; lag accumulates. DTN custody queue grows. CRDT merge frequency drops; drift increases. AI Supervisor tightens custody priority weights, elevates high-priority data classes, begins contact-plan optimization. Human operator receives lag-trend alerts; mode transition to Blackout is recommended, not automatic.

Blackout mode. Link absent. No synchronization; nodes operate independently. WAL continues to log locally; DTN bundles queue for the next contact window; CRDT state diverges under local writes. AI Supervisor enters custody-only posture — no parameter adjustments that could cause divergence between nodes. Human decision gate required for any policy change that affects both sides of the partition.

Mode transition authorization. The human-governs rule is clearest at mode transitions. The Supervisor models the transition, computes the expected state at reconnect, and presents the recommendation. The operator authorizes. This is the H half of HGC³AE² in its most operationally consequential form.

Unpacked in P6 and P7. Protocol selection within each mode (when to use CRDT vs. gossip, what custody priority schedule to apply) is the P6 problem. Workload-class-specific mode behavior (how a targeting workload behaves differently from a logistics workload in Blackout mode) is the P7 problem.

Figure 4 — Three Operational Modes *State-transition diagram showing Connected → Degraded → Blackout and the reverse transitions. Each mode annotated with WAL replication posture, CRDT merge frequency, DTN custody queue behavior, and AI Supervisor operating bounds. Human decision gates marked at each transition.* [FIGURE SEED: assets/series-3-edo/figures-seed/slide-10.png — full SVG redraw at v1.0-preprint]

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§7 — THE C5ISR APPLICATION LENS: FIVE WORKLOAD CLASSES, ONE ARCHITECTURE

Thesis: C5ISR applications at the tactical edge are not monolithic — they span five distinct workload classes with different latency tolerances, state-coherence requirements, and blackout behaviors. The substrate architecture is the same for all five; the mode-specific policies differ. One substrate, five policies.

Workload Class 1 — Situational Awareness (SA). Persistent state, moderate latency tolerance, high value of eventual consistency. SA data must converge after a blackout; stale SA is preferable to no SA. CRDT-native; gossip protocol acceptable in Degraded mode.

Workload Class 2 — Fire Control / Targeting. Time-critical, low-latency tolerance, strict ordering requirements. In Blackout mode, fire control data queues under DTN custody with highest priority weight. CRDT not appropriate for targeting solutions — WAL ordering and human confirmation required before commit.

Workload Class 3 — Intelligence / Sensor Feed. High volume, high value, acceptable loss rate on individual frames. Bulk transfer under DTN custody; partial delivery acceptable. CRDT applicable at the aggregated-product layer, not the raw-feed layer.

Workload Class 4 — Logistics / Sustainment. Low latency tolerance, high eventual-consistency value. Logistics data is a natural CRDT candidate — quantities, locations, and requests can be merged without coordinator. Blackout mode: queue and deliver; no human gate required for routine resupply data.

Workload Class 5 — Command and Control (C2). Orders, authorizations, and mission-coordination packets. Highest custody priority. Human authorization required for any modification to a C2 bundle in transit. Fail-closed default: if an authorization cannot be confirmed before a contact window closes, the bundle is held, not released.

Figure 5 — C5ISR Application Classes *Five-row table showing workload class, latency tolerance, state-coherence mechanism, CRDT applicability, and Blackout-mode behavior. Annotated with the substrate component that governs each class's behavior in each mode.* [FIGURE SEED: assets/series-3-edo/figures-seed/slide-11.png — full SVG redraw at v1.0-preprint]

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§8 — CLASSIFICATION, CYBER, INTEROP, OEM: NAMED, NOT UNPACKED

Thesis: Four cross-cutting concerns — classification surface management, cybersecurity posture, interoperability with legacy and allied systems, and OEM deployment reference — each represent a full problem-unpack paper. This section names them and points forward. A reader who needs the engineering detail goes directly to the indicated paper.

Classification surface (→ P8). The substrate handles multi-level security data. Bundle custody does not automatically enforce classification boundaries — MLS tagging, CDS passage, and accreditation of cross-domain flows are explicit engineering requirements. P8 (The Classification Problem) unpacks the surface: what data crosses what boundary, how tagging is applied and verified, and what the RMF control family looks like for a DTN-based MLS system. See slide 12 for the boundary taxonomy.

Cybersecurity posture (→ P9). Bundle signing authenticates custody hops — a bundle accepted by a DTN agent carries a verifiable signature chain from the originating node. Key management, certificate lifecycle, and RMF accreditation of the bundle-signing surface are P9 concerns (The Accreditation Problem). NIST SP 800-53 [@NIST800-53r5] and DFARS 252.204-7012 [@DFARS-252-204-7012] control families map to the signing and custody surfaces. See slide 14.

Interoperability (→ P10). The substrate must coexist with legacy military communication systems (Link 16, VMF, STANAG protocols) and allied partner networks operating under different DTN implementations. Protocol translation, gateway architecture, and the standards plug-in map are the P10 problem (The Interoperability Problem). See slide 15.

OEM reference stack and BOM (→ P11). Acquirers and program managers need a deployable reference stack: ruggedized NAS hardware, DTN software stack (NASA JPL ION [@NASA-ION]), WAL implementation, and the integration envelope. P11 (The Deployment Problem) provides the OEM reference architecture, bill of materials, and the integration testing framework. See slide 16.

Figures 12, 14, 15, and 16 from the v4 source deck are designated as the hero figures for P8, P9, P10, and P11 respectively. They are not reproduced here — the P-series paper for each concern carries the figure at full depth.

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§9 — READING GUIDE: WHICH PAPER TO READ NEXT

This paper is the survey frame. A reader who needs to act on the architecture — to deploy it, accredit it, procure it, or build on it — needs the problem-unpack paper for their specific concern. The table and prose below provide the directed reading path for four reader roles.

Reader Role	Start Here	Then
Operator / Warfighter	§6 (Three Modes) in this paper	P6 (Protocol Selection) → P7 (Workload Class)
Systems Architect	This paper, cover to cover	P3 (Transport) → P4 (State Coherence) → P5 (Orchestration) → P6 (Protocol Selection)
Accreditor / RMF Reviewer	§8 of this paper	P8 (Classification) → P9 (Accreditation)
Acquirer / Program Manager	§8 of this paper	P10 (Interoperability) → P11 (Deployment)

Operator / Warfighter. The three operational modes in §6 of this paper are the operational vocabulary the rest of the series uses. Start there — you should be able to name what changes between Connected, Degraded, and Blackout mode before reading anything else. P6 (Protocol Selection) then gives you the decision logic for when CRDT vs. gossip applies — this is the operationally consequential protocol choice because it determines how long state can safely diverge before a merge is required. P7 (Workload Class) gives you the per-mission-type policy: how a targeting workload behaves differently from a logistics workload in Blackout mode, and what the human-authorization requirements look like for each.

Systems Architect. Read this paper end to end — it is the only paper in the series that provides the full substrate overview in one place. Then sequence through the problem-unpack papers in dependency order: P3 (Transport Problem) for the DTN implementation detail and contact-plan engineering; P4 (State Coherence Problem) for the WAL-on-NAS topology, CRDT selection criteria, and the merge-after-blackout walkthrough; P5 (Orchestration Problem) for the AI Supervisor design — event taxonomy, escalation protocol, and control-plane interface; and P6 (Protocol Selection) for the decision framework that governs which synchronization mechanism applies in which operational mode. The architect who has read P3–P6 can build the substrate. P7–P11 are the application-domain and compliance engineering layers.

Accreditor / RMF Reviewer. Section §8 of this paper names the four accreditation surfaces — classification, cybersecurity, interoperability, and deployment — without unpacking them. Your first read is §8; it gives you the surface map and the paper pointers. P8 (The Classification Problem) is the next read: it provides the MLS surface taxonomy, the CDS passage requirements, and the RMF control family mapping for DTN-based classification routing. P9 (The Accreditation Problem) follows: it provides the bundle-signing accreditation surface, the NIST SP 800-53 control mapping, and the DFARS 252.204-7012 [@DFARS-252-204-7012] obligation analysis for OEM vendors deploying the substrate.

Acquirer / Program Manager. Section §8 of this paper gives you the four-sentence version of each cross-cutting concern and a pointer to the paper that unpacks it. Your directed read after §8 is P10 (The Interoperability Problem) for the standards plug-in map — this tells you what standards the substrate implements, what legacy systems it can gateway with, and what allied interoperability constraints the procurement must accommodate. P11 (The Deployment Problem) gives you the OEM reference stack, bill of materials, and integration testing framework — this is the document your program office hands to a system integrator.

P2 — HGC³AE² at the Degraded Edge — is the governor paper. Every reader should pull it alongside their specialty paper; it defines the human-governance structure that every component of the substrate operates inside.

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§9.5 — GOVERNOR LENS: HGC³AE² ACROSS THE SUBSTRATE

Stub — full treatment in P2 "HGC³AE² at the Degraded Edge."

The HGC³AE² framework [@HGC3AE2-S1P1] was developed in Series 1 as the engineering grammar for human-governed agentic systems. *The Implications of Edge Degraded Ops* applies that grammar to the specific problem of distributed state synchronization under DDIL conditions. The application is not metaphorical — the H / C³ / AE² split maps directly onto the substrate architecture, and the mapping is load-bearing: violation of the ordering (humans govern, context is curated, AI executes) produces failure modes that are both technical and operational.

The **H half** (Human Governance) governs the substrate's policy surface. Humans set drift thresholds — the maximum CRDT divergence acceptable before a merge is required. Humans approve classification routing changes — a decision to permit a class of data to cross a boundary that was previously closed. Humans decide when Blackout mode triggers custody-only posture, which determines what the AI Supervisor is and is not permitted to adjust during the partition. The AI Supervisor does not govern — it executes. When the Supervisor's model indicates that a threshold is about to be breached, it alerts; the human decides. This is not a performance constraint — it is the architecture's compliance with the operational requirement that lethal and sensitive decisions remain in the human chain of command.

The **C^3 half** (Curation, Cataloging, Curriculum) maps onto the data lifecycle of the substrate. Curation is the determination of what gets custody-prioritized in Degraded mode — when the contact window is short and the queue is long, the curation policy determines what ships. That policy is human-set and AI-assisted, not AI-determined. Cataloging is the WAL: every state transition is logged with enough fidelity to reconstruct provenance, support audit, and drive the CRDT merge operation. The WAL is not just a technical mechanism — it is the catalog the accreditor reads. Curriculum is the AI Supervisor's model of link behavior: the contact-plan priors, the link-quality history, the merge-outcome patterns that inform the Supervisor's parameter adjustments. That model is updated from operational history, reviewed by human operators, and can be rolled back.

The **AE^2 half** (Authentication and Enforcement) governs the trust surface of every custody hop and every state boundary. Bundle signing authenticates custody transfers — a DTN agent does not accept custody of a bundle it cannot verify. MLS tags enforce classification boundaries — data does not cross a classification boundary without the CDS passage that authorizes it. Fail-closed is the default: when the AE^2 enforcement surface cannot verify a claim — a signature it cannot validate, a classification tag it cannot authorize — the data does not move. No unclassified data crosses an accredited boundary without explicit CDS passage. No custody acknowledgment is issued for a bundle that fails authentication. The AE^2 half does not negotiate; it enforces.

The full treatment of HGC^{3}AE^2 applied to this substrate is in P2 — "HGC^{3}AE^2 at the Degraded Edge." Every problem-unpack paper (P3–P11) carries a §6 Governor Application section that applies this lens at its specific problem scale.

BIBLIOGRAPHY

Citation keys correspond to assets/series-3-edo/series-3-citations.bib. Import that file into any BibTeX render of this paper. Internal series cites marked as forthcoming where the paper is not yet published.

Load-Bearing (□□□)

[@HGC3AE2-S1P1] Yeti. *HGC^{3}AE^2 — Human Governance, Curation^3, and AI Enforcement^2 as an Integrated Framework for Agentic Operations.* nonsequitur.tech, 2025. *P1 load-bearing parent. The H / C^3 / AE^2 engineering grammar underlies every section of this paper. §9.5 is the first application of that grammar to the tactical substrate; the full application is in P2.*

[@EdgeAIDoctrine-S1P3] Yeti. *Edge AI Doctrine — Inference at the Tactical Edge: Governance, Constraint, and the Limits of Autonomy.* nonsequitur.tech, 2025. *Direct predecessor. This series extends inference-at-edge to state-sync-at-edge. The AI Supervisor pattern (§5) is the state-sync counterpart to the edge inference governance model in this paper.*

[@OpDiscipline-S1P2] Yeti. *Operational Discipline Doctrine — Five Deliberate-Stop Classes for Human-Governed Agentic Systems.* nonsequitur.tech, 2025. *The deliberate-stop taxonomy maps directly onto the three-mode envelope. Blackout mode is the maximum-constraint deliberate-stop posture. The AI Supervisor's fail-closed default is the operational discipline the stop classes require.*

Thesis-Supporting (□□)

[@Skipjack-S1Cap] Yeti. *The Skipjack Protocol — Operational Discipline for Agentic Systems Under Constraint.* nonsequitur.tech, 2025. *This series lives inside the Skipjack operational discipline frame. The five deliberate-stop classes map to the three-mode envelope; Skipjack's custody-first posture is the model for DTN bundle custody under constraint.*

[@RFC5050] Scott, Keith L. and Burleigh, Scott. *Bundle Protocol Specification.* RFC 5050. Internet Engineering Task Force, 2007. https://www.rfc-editor.org/rfc/rfc5050 *The DTN bundle protocol specification. The custody mechanism described in §3 and §4 is RFC 5050 bundle custody. Referenced at survey depth here; load-bearing in P3 (Transport Problem).*

[@NASA-ION] NASA Jet Propulsion Laboratory. *ION (Interplanetary Overlay Network) — DTN Reference Implementation.* NASA JPL Open Source, 2023. https://ion.nasa.gov/ *The reference implementation of RFC 5050 bundle protocol. The concrete software anchor for P3 and P11.*

Standards and Compliance

[@NIST800-53r5] Joint Task Force. *Security and Privacy Controls for Information Systems and Organizations.* NIST Special Publication 800-53 Rev. 5. National Institute of Standards and Technology, 2020. https://doi.org/10.6028/NIST.SP.800-53r5 *RMF control family reference for the AE^2 enforcement surfaces named in §8 and §9.5. Load-bearing in P9 (Accreditation Problem).*

[@DFARS-252-204-7012] Department of Defense. *Safeguarding Covered Defense Information and Cyber Incident Reporting.* Defense Federal Acquisition Regulation Supplement, 48 CFR 252.204-7012. *DoD contractor cybersecurity obligation. Referenced in §8; load-bearing in P8 (Classification Problem) and P9 (Accreditation Problem).*

The Implications of Edge Degraded Ops — Forthcoming

The following papers are planned deliverables in this series. They are forward-referenced throughout this paper and will be registered with permanent DOIs on Zenodo deposit.

[@S3P2-HGC3AE2DegradedEdge] Yeti. *HGC^3AE^2 at the Degraded Edge. The Implications of Edge Degraded Ops*, P2. nonsequitur.tech, 2026. *Forthcoming.*

P3 — *The Transport Problem. The Implications of Edge Degraded Ops*, P3. *Forthcoming.*

P4 — *The State Coherence Problem. The Implications of Edge Degraded Ops*, P4. *Forthcoming.*

P5 — *The Orchestration Problem. The Implications of Edge Degraded Ops*, P5. *Forthcoming.*

P6 — *The Protocol Selection Problem. The Implications of Edge Degraded Ops*, P6. *Forthcoming.*

P7 — *The Workload Class Problem. The Implications of Edge Degraded Ops*, P7. *Forthcoming.*

P8 — *The Classification Problem. The Implications of Edge Degraded Ops*, P8. *Forthcoming.*

P9 — *The Accreditation Problem. The Implications of Edge Degraded Ops*, P9. *Forthcoming.*

P10 — *The Interoperability Problem. The Implications of Edge Degraded Ops*, P10. *Forthcoming.*

P11 — *The Deployment Problem. The Implications of Edge Degraded Ops*, P11. *Forthcoming.*

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v0.1-seed — §1, §9, §9.5 drafted in full; §2–§8 at 1-para thesis + 3–5 bullet outline per v0.1-seed posture. Full prose at v1.0-preprint. nonsequitur.tech review precedes Zenodo deposit.